

series_blueprint

Preprint Series Blueprint

Series title: **Spatium Computationis v3: Physical Substrates for Autonomous Intelligence**

Series authority state: INTERNAL_RESEARCH

Publication posture: partial_repo_verified and needs citation hardening

Repository inspected: <https://github.com/FreddyCreates/pegasus-battleops>

Inspected commit: a756fa39e39342dc2443c6c7d9335f4c9d68455c

Repository verification state: repo_verified_architecture_partial_engine_gap

Series Thesis

Spatium Computationis v3 proposes that autonomous intelligence systems can be modeled as agents embedded in a physically constrained computational substrate, where simulation dynamics, differential geometry, cosmological sandboxing, and spatial agent routing become separable but connected layers of system design.

Recommended Series Structure

Paper 1: Numerical Mechanics Substrate

Title: **A Symplectic Physics Substrate for Autonomous Agent Simulation**

Distinct thesis: A physics substrate using Velocity Verlet integration, N-body interactions, relativistic speed bounds, elastic collisions, and conservation diagnostics can provide a bounded numerical environment for modeling agents as physical entities.

Primary contribution: numerical mechanics and validation framework.

Do not cover in depth: curvature tensors, mini-verse cosmology, agent battleground taxonomy.

Paper 2: Riemannian Geometry Engine

Title: **Numerical Riemannian Geometry for Runtime-Modifiable Computational Spacetimes**

Distinct thesis: A metric-function-first geometry engine can compute Christoffel symbols, geodesics, curvature tensors, horizons, and parallel transport numerically, enabling runtime-modifiable spacetime models without requiring closed-form symbolic derivations.

Primary contribution: computational differential geometry engine.

Do not cover in depth: N-body force laws, cosmological topology catalog, threat-routing architecture.

Paper 3: Mini-Verse Cosmology and Topological Sandboxes

Title: **Mini-Verse Sandboxes: Configurable Cosmological Substrates for Intelligence Experiments**

Distinct thesis: Isolated mini-verses with configurable constants, topologies, Friedmann evolution, entropy accounting, and bounded inter-verse channels can serve as experimental substrates for studying agent behavior under varied physical laws.

Primary contribution: universe-level sandbox architecture.

Do not cover in depth: integrator derivations, full curvature implementation, 21-agent spatial taxonomy.

Paper 4: Agent Embedding and Battleground Architecture

Title: **Embedding Autonomous Agents in Simulated Spacetime: A Battleground Architecture for Routing, Isolation, and Coordination**

Distinct thesis: Spatially embedding agents and zones inside a simulated substrate allows routing, quarantine, coordination, and influence to be expressed as geometric and physical relationships rather than only abstract software calls.

Primary contribution: autonomous agent architecture and spatial operational model.

Do not cover in depth: full numerical physics validation, tensor derivations, cosmological Friedmann analysis.

Publication Order

1. Paper 1 establishes the numerical substrate and validation norms.
2. Paper 2 extends the substrate into geometric computation.
3. Paper 3 packages geometry and physics into isolated universe-like experimental domains.
4. Paper 4 applies the substrate to autonomous intelligence architecture.

Shared Lineage Rule

Each paper should cite the canonical packet as the internal source lineage but should publish only its own contribution surface. Shared introduction text should be limited to one short paragraph defining Spatium Computationis and one paragraph stating the paper's position in the series.

Public/Private Boundary

Public-safe:

- Mathematical definitions of metrics, geodesics, conservation laws, and cosmological equations.
- General claim that the architecture proposes physically grounded agent simulation.
- High-level mini-verse taxonomy and sandbox framing.

Needs evidence before public promotion:

- Production-grade implementation claims.
- Quantitative validation thresholds such as energy error below 10^{-6} .
- Performance claims such as suitability for $N < 10000$.

IP/release review first:

- Detailed battleground routing logic.
- Security-oriented tiering and adversary-lab semantics.
- Exact agent placements if treated as strategic operational architecture.

Series-Level Evidence Needed

- Repository snapshot or release archive from `FreddyCreates/pegasus-battleops`.
- File-level evidence for Velocity Verlet/N-body physics, Riemannian geometry, and mini-verse cosmology if those remain paper claims.
- Commit hash matching the described implementation.
- Test logs for conservation laws, geodesic norm preservation, horizon detection, and topology behaviors.
- Benchmark table for N-body scaling, geodesic computation, curvature computation, and Web-Socket simulation tick behavior.
- Reproducibility script or notebook for at least one figure/table per paper.
- Citation pass for differential geometry, symplectic integration, cosmology, agent-based modeling, and cyber/defense architecture analogies.